P425/2

PAPER 2

APRIL - 2026

2HOURS: 45MINS

ST JAMES SECONDARY SCHOOL-KITAGOBWA
S.6 END OF TERM 1

Uganda Advanced Certificate of Education

ASSESMENTS 2026.

Paper 2

TIME: 2HOURS 45MINS

INSTRUCTIONS TO LEARNERS:

1. *Answer **all** questions in this paper in section A.*
2. *Choose **Any Four (4) questions** from all sections, atleast choosing two from each section*
3. *All answers should be written in **blue** or **black** ink in the answer sheets provided. (Graph papers are also provided.)*
4. *All necessary working must be shown clearly.*
5. *Silent, non-programmable scientific calculators may be used.*

ITEMS	For Examiner's Use Only				
	Marks	**Initials**	**ITEMS**	**Marks**	**Initials**
1			**5**		
2			**6**		
3			**7**		
4			**TOTAL (100%)**		

CLEARLY INDICATE THE ITEM NUMBERS ATTEMPED ON YOUR FIRST PAGE

SECTION A

(Choose any two Items from this section)

ITEM 1

You are selling you land and the council tasked you to bring a surveyor for measuring before selling. A surveyor uses a measuring tape with errors cause by expansion as the length and width of 5% and 4.2% errors respectively in estimates. Your land has dimensions of a rectangular plot as 10.25km by 6.44km. Each Km^2 of your dad's land is to be sold at 12.5M. After selling, the money is deposited at the bank but your dad keeps on lamenting that he was stolen.

The numbers $\boldsymbol{a}$ and $\boldsymbol{b}$ are approximated with errors of $\boldsymbol{e_1}$ and $\boldsymbol{e_2}$ respectively. Given that $\boldsymbol{a=3.3214}$ and $\boldsymbol{b=6.325}$

TASK

a) Advise your dad of the money range he should expect on the account.

b) (i). Show that the maximum relative error in the quotient $\frac{a^2}{b}$ is given by

$$2\left|\frac{e_1}{a}\right|+\left|\frac{e_2}{b}\right|.$$

(ii). Find the limits where $\frac{a^2}{b}$ is expected to lie

ITEM 2

The coffee farmer bought coffee seeds of three varieties of seeds S1, S2 and S3 in the ratio 4:2:1. The research shows that the probabilities of germination of S1 S2 and S3 are 50%, 60% and 80% respectively.

A family is planning to have 6 Children according to the available resources. And according the health records of their Potential, a probability od producing a girl is twice that of getting a boy. And a society norm, each boy is always given a land as a starting capital when he marries. A family head is not well sure whether his land is enough to cater for the boys that may arise in his family.

TASK

(a) (i) Find the probability that a seed selected at random will germinate.

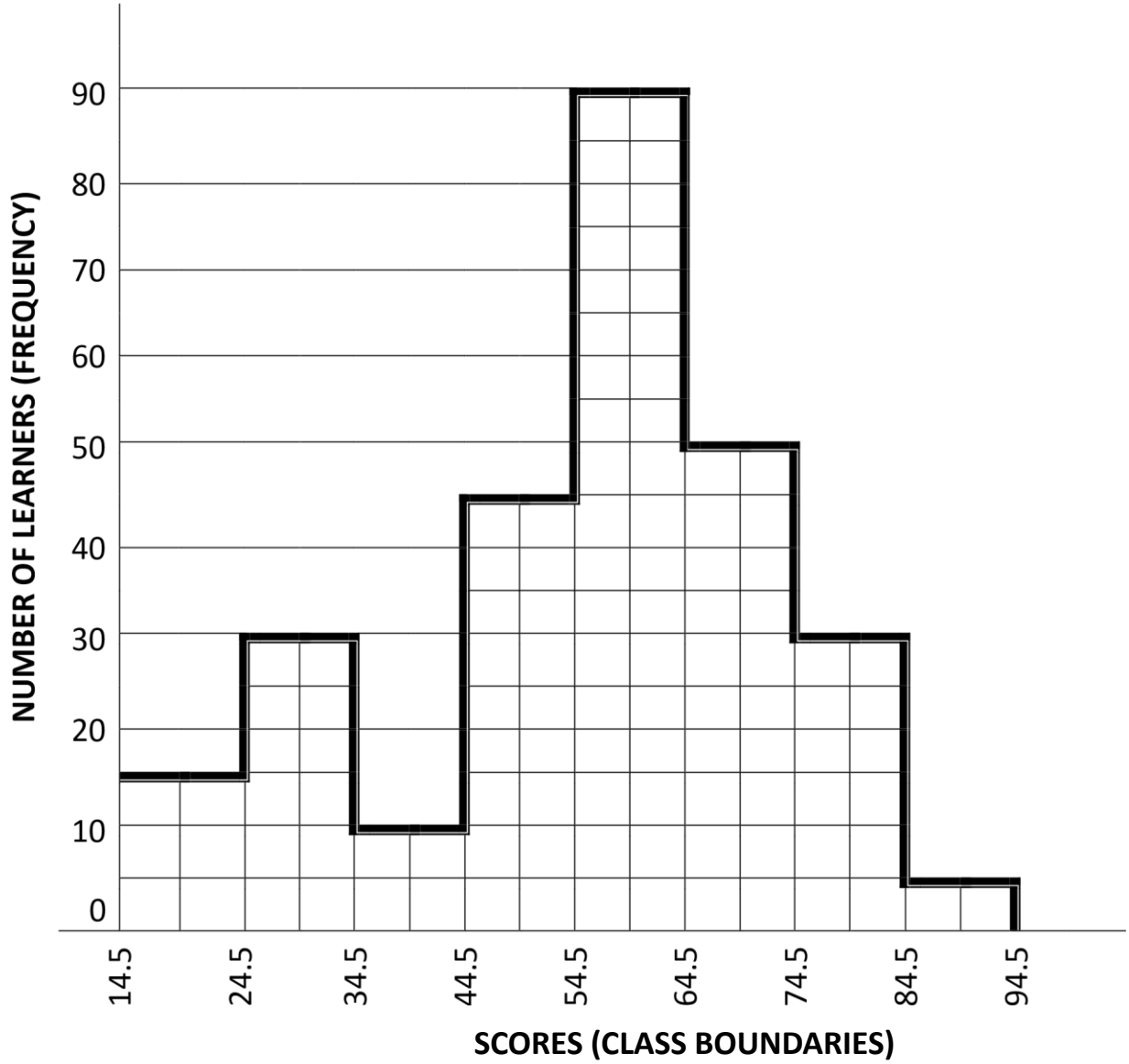
(ii) Given that 150 seeds were picked, find the probability that less than 90 of the seeds will germinate. Give your answer to two decimal places.

b) Advise the Family head of the expected number of boys to assist him in planning

ITEM 3

A certain school employed a new teacher of mathematics to help the Senior Six leaners to improve on their performance. Before the new teacher was employed, analysis showed that the learners' mean score was 54% and 38% of the leaners scored below the pass mark of 55%. The new teacher has taught for full term and leaners have been given an exam whose results are shown in graph one below

GRAPH SHOWING THE NUMBER OF LEANERS AND THEIR SCORES IN MATHEMATICS



The school needs help to analyse the results of leaners performance shown on the graph above and establish whether there is an improvement in the learners' performance basing on

(i) The mean score of the results on graph 1 in comparison to the previous mean score.

(ii) The percentage of leaners below the school pass mark of 55%

TASK

Provide the school with the help they need.

ITEM 4

A factory produces a high precision circuit boards. Due to complexity of the assembly process, it is known from historical data that 15% of the boards produced have a minor soldering defect. The quality control team randomly selects a batch of 20 boards for inspection before shipping

The acceleration response of these vehicles towards the traffic's passenger regulations

,

rule along high way check points is described by

$$f(x) = \begin{cases} 2k(x+1), & -1 \le x \le 0 \\ k(2-x), & 0 \le x \le 2 \\ 0, & elsewhere \end{cases} ,$$

Most drivers have been taken to court but later found guilty of the offences which have caused the traffic police to compensate much money to these drivers. The police approached you as a student of mathematics to aid to find the acceleration any kind of acceleration variance if possible

(a) What is the chance that

i) Exactly 3 boards in the batch are defective?

ii) At most 4 boards are defective

(b) Probability that at least 11 vehicles leave the stage

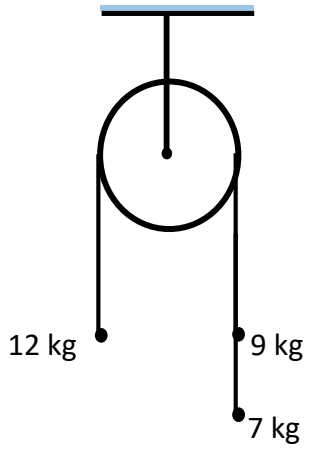
(c) Help the police of the problems at hand to reduce the risks of compensation to drivers found guilty of offenses.

SECTION B

(Choose any two Items from this section)

ITEM 5

During Physics practical exam, different masses were put on the pulley system by students. The diagram below shows three masses of 12 kg, 9kg, and 7kg connected by light inextensible strings. The string connecting the 12kg and the 9kg masses pass over a smooth fixed pulley. The other string connects the 9kg and 7kg masses. As shown below.



The system is released from rest and the 12 mass accelerates upwards.

(a). Calculate the;

(i). acceleration of the system.

(ii). tensions in the strings.

ITEM 6

During the Construction development plans, more calculations are put in place considering the directions of the wind. The building was expected to have vertices ABCDEF is a regular Hexagon. Wind Forces of Magnitude 3N, 4N, 2N, and 6N act along the sides AB, AC, EA, and AF respectively. In each case the direction of the force being given by the order of letters. AB and AE are the horizontal and vertical axis respectively. If this wind is not calculated for in the foundation plans, the building may fall so to avoid, a strong resistant concrete column is to be put in place to oppose the force.

WXYZ is a rectangle with sides WX = 4m and XY = 3m. Forces of magnitude 3N, 10N, 4N, 6N, and 5N act along the line WX, XY, YZ, ZW, and WY respectively, in each case the direction of force being given by the order of the letters.

TASK

a) In which force and direction should the column be positioned to make the building in equilibrium to avoid collapse.

b) Taking WX as horizontal axis, determine the magnitude and direction of the resultant force.

ITEM 7

Two wooden bars in a school compound were affected by forces to change the position.

Wood 1; The forces of 2N, 3N, 4N, 5N and 6N acted up on a wooden bar of 450g in a

school compound from directions of 030^0, 090^0, 120^0, 210^0, and 330^0 respectively. **Wood 2;** Forces of 6N, 5N, 7N, 8N, 5N and 9N act on this wood in the directions $N30^0E$, $N30^0W$, $S50^0E$, $N60^0W$, $N60^0E$ and $S40^0W$ respectively.

TASK

a) What is the acceleration of wood one

b) Derive one driving Force that would do the same work on wood 2

ITEM 8

During the lesson, a teacher stated the equations of Uniform acceleration (equations of motion) but deep in minds of students failed to appear as expected. More of students kept doubting the equations which calls for doubts in teachers by the administration. A car being driven along a road at a steady speed of $25ms^{-1}$, when suddenly the driver notices a fallen tree on the road 65m a head. The driver immediately applies brakes giving the car a constant retardation of $5ms^{-2}$

a) Help your class to come at the derived equations of motion.

(b) i) How far in front of a tree does the car come to rest

ii) If the driver had not reached immediately and the breaks were applied one second later, with what speed would the car have hit the tree?

END